UNIT 2 · TOPIC 10(D)

Robotics and Automation

Syllabus topic: Machine learning, IoT, robotics and mechatronics in farm machinery

Robot elements

Element	Role
Manipulator	Links + joints (degrees of freedom)
End-effector	Gripper/cutting tool that contacts the product
Sensors	Machine vision, LiDAR, GPS, encoders
Controller	Path planning

Field examples

- RTK-GNSS auto-steer tractors (**2–3 cm** pass-to-pass accuracy) and fully autonomous tractors.
- Intra-row robotic weeders (vision + hoe or laser).
- Robotic transplanters.
- Fruit/vegetable harvesting robots — vision locates the produce; a soft-gripper end-effector detaches it.
- Robotic milking parlours; greenhouse spray robots.

Design issues for agri-robots

Unstructured outdoor environment (light, dust, mud); biological variability of produce; gentle handling (bruising); safety around people; power/endurance; cost-effectiveness versus scarce labour.

Practice MCQs

Q201. RTK-GNSS auto-steer gives pass-to-pass accuracy of about:

(a) 2–3 m (b) 30–50 cm
(c) 2–3 cm (d) 2–3 mm

Q202. The robot component that actually grips/cuts the fruit is the:

(a) Manipulator base (b) End-effector
(c) Encoder (d) Gantry

Answers

Q201 — (c) 2–3 cm
Q202 — (b) End-effector

Sources: Kepner et al. (hydraulic principles); Drone Rules 2021 (Ministry of Civil Aviation); MoA&FW SOPs for drone application of pesticides & nutrients; SMAM drone guidelines & PIB releases (Kisan Drone, Namo Drone Didi — verified July 2026); ISO 11783 (ISOBUS); NPTEL lectures on mechatronics & precision agriculture.